

Your grade: 100%

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Next item →

1. Select all of the statements, below, which are true.

1 / 1 point

☒ Joint estimation uses a single nonlinear Kalman filter to estimate both system state and system parameters via a combined state/parameter vector.

☒ **Correct**
Yes. This is true.

☐ Dual estimation uses a single nonlinear Kalman filter to estimate both system state and system parameters via a combined state/parameter vector.

☐ Joint estimation uses two separate nonlinear Kalman filters to estimate both system state and system parameters via separate state and parameter vectors.

☒ Dual estimation uses two separate nonlinear Kalman filters to estimate both system state and system parameters via separate state and parameter vectors.

☒ **Correct**
Yes. This is true.

2. What are advantages of joint estimation over dual estimation? (Select all that apply.)

1 / 1 point

☐ Joint estimation is less computationally complex (requires fewer floating-point operations per iteration) due to smaller matrix operations.

☒ Joint estimation is more straightforward to implement. It is essentially the same algorithm used for nonlinear state estimation, but uses a larger "state" vector.

☒ **Correct**
Yes. This is true.

☒ Joint estimation may give better estimates since it does not ignore possible correlations between state and parameter errors.

☒ **Correct**
Yes. This is true.

☐ Joint estimation may be implemented using SPKF or EKF, but dual estimation may be implemented only using EKF.

☐ Joint estimation is better conditioned numerically due to how it handles different timescales of parameter and state variations.

3. What are advantages of dual estimation over joint estimation? (Select all that apply.)

1 / 1 point

☐ Dual estimation is more straightforward to implement. It is essentially the same algorithm used for nonlinear state estimation, but uses a larger "state" vector.

☒ Dual estimation is less computationally complex (requires fewer floating-point operations per iteration) due to smaller matrix operations.

☒ **Correct**
Yes. This is true.

☒ Dual estimation is better conditioned numerically due to how it handles different timescales of parameter and state variations.

☒ **Correct**
Yes. This is true.

☐ Dual estimation may give better estimates since it does not ignore possible correlations between state and parameter errors.

☐ Dual estimation may be implemented using SPKF or EKF, but joint estimation may be implemented only using EKF.